

ООП 2

ПОВЕДЕНЧЕСКИ ШАБЛОНИ ЗА ДИЗАЙН 3 ЧАСТ

ПРОБЛЕМ

- НАБОР ОТ ОБЕКТИ
 - НЕОБХОДИМОСТ ОТ НОВИ РАЗШИРЕНИ ФУНКЦИОНАЛНОСТИ
 - ДОБАВЯНЕ В КЛАСОВЕТЕ?

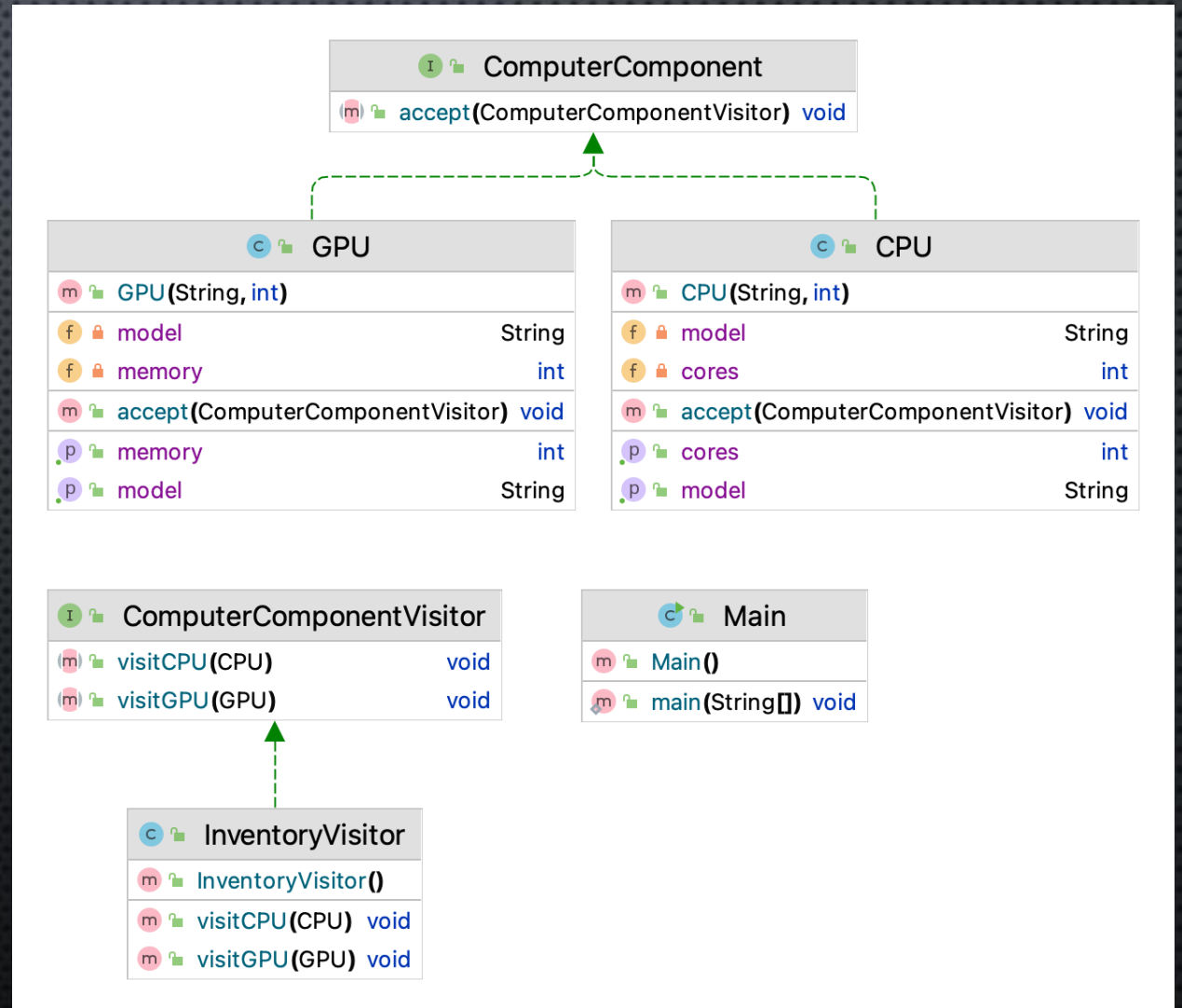
VISITOR ШАБЛОН

- ИЗНАСЯНЕ НА НОВАТА/ТЕ ФУНКЦИОНАЛНОСТ/И ВЪВ НОВ ОБЕКТ (VISITOR), КОЙТО СИ ВЗАИМОДЕЙСТВА С ВЕЧЕ ДЕФИНИРАНИТЕ КЛАСОВЕ

ПРОБЛЕМ

- КОМПЮТЪРНИ КОМПОНЕНТИ
 - ДОПЪЛНИТЕЛНИ ДЕЙСТВИЯ С ТЯХ

VISITOR ШАБЛОН



VISITOR ШАБЛОН

```
PUBLIC INTERFACE COMPUTERCOMPONENT {  
    VOID ACCEPT(COMPUTERCOMPONENTVISITOR VISITOR);  
}
```

VISITOR ШАБЛОН

```
PUBLIC CLASS CPU IMPLEMENTS COMPUTERCOMPONENT {
```

```
    PRIVATE STRING MODEL;
```

```
    PRIVATE INT CORES;
```

```
    PUBLIC CPU(STRING MODEL, INT CORES) {
```

```
        THIS.MODEL = MODEL;
```

```
        THIS.CORES = CORES;
```

```
    }
```

```
    PUBLIC STRING GETMODEL() {
```

```
        RETURN MODEL;
```

```
    }
```

```
    PUBLIC INT GETCORES() {
```

```
        RETURN CORES;
```

```
    }
```

```
    @Override
```

```
    PUBLIC VOID ACCEPT(COMPUTERCOMPONENTVISITOR  
VISITOR) {
```

```
        VISITOR.VISITCPU(THIS);
```

```
    }
```

```
}
```

VISITOR ШАБЛОН

```
PUBLIC CLASS GPU IMPLEMENTS COMPUTERCOMPONENT {
```

```
    PRIVATE STRING MODEL;
```

```
    PRIVATE INT MEMORY;
```

```
    PUBLIC GPU(STRING MODEL, INT MEMORY) {
```

```
        THIS.MODEL = MODEL;
```

```
        THIS.MEMORY = MEMORY;
```

```
    }
```

```
    PUBLIC STRING GETMODEL() {
```

```
        RETURN MODEL;
```

```
    }
```

```
    PUBLIC INT GETMEMORY() {
```

```
        RETURN MEMORY;
```

```
    }
```

```
    @OVERRIDE
```

```
    PUBLIC VOID ACCEPT(COMPUTERCOMPONENTVISITOR  
VISITOR) {
```

```
        VISITOR.VISITGPU(THIS);
```

```
    }
```

```
}
```

VISITOR ШАБЛОН

```
PUBLIC INTERFACE COMPUTERCOMPONENTVISITOR {  
    VOID VISITCPU(CPU CPU);  
    VOID VISITGPU(GPU GPU);  
}
```

VISITOR ШАБЛОН

```
PUBLIC CLASS INVENTORYVISITOR IMPLEMENTS  
COMPUTERCOMPONENTVISITOR {
```

```
    @Override
```

```
    PUBLIC VOID VISITCPU(CPU CPU) {
```

```
        SYSTEM.OUT.PRINTLN("CPU MODEL: " + CPU.GETMODEL());
```

```
        SYSTEM.OUT.PRINTLN("NUMBER OF CORES: " +  
CPU.GETCORES());
```

```
        // ДОПЪЛНИТЕЛНИ ОПЕРАЦИИ
```

```
    }
```

```
    @Override
```

```
    PUBLIC VOID VISITGPU(GPU GPU) {
```

```
        SYSTEM.OUT.PRINTLN("GPU MODEL: " + GPU.GETMODEL());
```

```
        SYSTEM.OUT.PRINTLN("MEMORY: " + GPU.GETMEMORY() + "  
GB");
```

```
        // ДОПЪЛНИТЕЛНИ ОПЕРАЦИИ
```

```
    }
```

```
}
```

VISITOR ШАБЛОН

```
PUBLIC CLASS MAIN {  
    PUBLIC STATIC VOID MAIN(STRING[] ARGS) {  
        CPU CPU = NEW CPU("INTEL CORE I7", 20);  
        GPU GPU = NEW GPU("NVIDIA GeForce RTX 4090", 24);  
        COMPUTERCOMPONENTVISITOR VISITOR = NEW INVENTORYVISITOR();  
        CPU.ACCEPT(VISITOR);  
        SYSTEM.OUT.PRINTLN("-----");  
        GPU.ACCEPT(VISITOR);  
    }  
}
```

VISITOR ШАБЛОН

CPU MODEL: INTEL CORE I7

NUMBER OF CORES: 20

GPU MODEL: NVIDIA GEFORCE RTX 4090

MEMORY: 24 GB

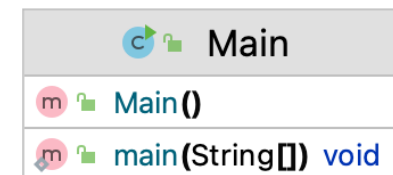
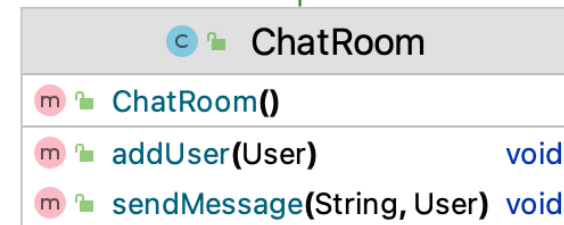
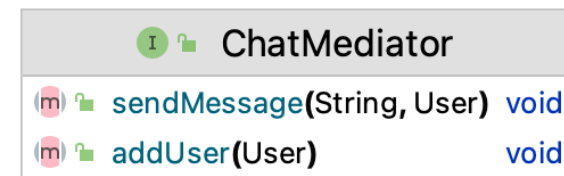
VISITOR ШАБЛОН

- РАЗДЕЛЯНЕ НА ОТГОВОРНОСТ
 - VISITOR КЛАСА КАПСУЛИРА В СЕБЕ СИ НОВИТЕ ФУНКЦИОНАЛНОСТИ
- ПОДОБРЯВА МОДУЛНОСТТА НА КОДА И ПОДДРЪЖКАТА
- SINGLE RESPONSIBILITY
- ИЗБЯГВАНЕ НА МОДИФИЦИРАНЕ НА СЪЩЕСТВУВАЩ КОД
 - ЧУЖД КОД!
- ГЪВКАВ НАЧИН ЗА ДЕФИНИРАНЕ НА МНОЖЕСТВО АЛГОРИТМИ ИЛИ ПОВЕДЕНИЯ ЗА НАБОР ОТ КЛАСОВЕ

ПРОБЛЕМ

- НЕОБХОДИМОСТ ОБЕКТИ ДА КОМУНИКИРАТ ПО МЕЖДУ СИ

MEDIATOR ШАБЛОН



MEDIATOR ШАБЛОН

```
PUBLIC INTERFACE CHATMEDIATOR {  
    VOID SENDMESSAGE(STRING MESSAGE, USER SENDER);  
    VOID ADDUSER(USER USER);  
}
```

MEDIATOR ШАБЛОН

```
PUBLIC CLASS User {  
    PRIVATE STRING NAME;  
    PRIVATE CHATMEDIATOR CHATMEDIATOR;  
  
    PUBLIC User(STRING NAME, CHATMEDIATOR CHATMEDIATOR) {  
        THIS.NAME = NAME;  
        THIS.CHATMEDIATOR = CHATMEDIATOR;  
    }  
}
```

```
PUBLIC VOID SENDMESSAGE(STRING MESSAGE) {  
    CHATMEDIATOR.SENDMESSAGE(MESSAGE, THIS);  
}  
  
PUBLIC VOID RECEIVEMESSAGE(STRING MESSAGE) {  
    SYSTEM.OUT.PRINTLN(NAME + " RECEIVED MESSAGE: " +  
MESSAGE);  
}  
}
```

MEDIATOR ШАБЛОН

```
IMPORT JAVA.UTIL.ARRAYLIST;
```

```
IMPORT JAVA.UTIL.LIST;
```

```
PUBLIC CLASS CHATROOM IMPLEMENTS CHATMEDIATOR {
```

```
    PRIVATE LIST<USER> USERS;
```

```
    PUBLIC CHATROOM() {
```

```
        THIS.USERS = NEW ARRAYLIST<>();
```

```
    }
```

```
    @OVERRIDE
```

```
    PUBLIC VOID SENDMESSAGE(STRING MESSAGE, USER SENDER) }
```

```
{
```

```
    FOR (USER USER : USERS) {
```

```
        IF (USER != SENDER) {
```

```
            USER.RECEIVEMESSAGE(MESSAGE);
```

```
        }
```

```
    }
```

```
}
```

```
@OVERRIDE
```

```
PUBLIC VOID ADDUSER(USER USER) {
```

```
    USERS.ADD(USER);
```

```
}
```

MEDIATOR ШАБЛОН

```
PUBLIC CLASS MAIN {  
    PUBLIC STATIC VOID MAIN(STRING[] ARGS) {  
        CHATMEDIATOR CHATMEDIATOR = NEW  
        CHATROOM();  
  
        USER USER1 = NEW USER("OBI_WAN",  
        CHATMEDIATOR);  
        USER USER2 = NEW USER("YODA", CHATMEDIATOR);  
        USER USER3 = NEW USER("LUKE", CHATMEDIATOR);
```

```
        CHATMEDIATOR.ADDUSER(USER1);  
        CHATMEDIATOR.ADDUSER(USER2);  
        CHATMEDIATOR.ADDUSER(USER3);  
  
        USER1.SENDMESSAGE("HELLO, THERE!");  
        USER2.SENDMESSAGE("POWERFUL YOU HAVE  
        BECOME!");  
        USER3.SENDMESSAGE("I AM A JEDI, LIKE MY  
        FATHER!");  
    }  
}
```

MEDIATOR ШАБЛОН

YODA RECEIVED MESSAGE: HELLO, THERE!

LUKE RECEIVED MESSAGE: HELLO, THERE!

OBI_WAN RECEIVED MESSAGE: POWERFUL YOU HAVE BECOME!

LUKE RECEIVED MESSAGE: POWERFUL YOU HAVE BECOME!

OBI_WAN RECEIVED MESSAGE: I AM A JEDI, LIKE MY FATHER!

YODA RECEIVED MESSAGE: I AM A JEDI, LIKE MY FATHER!

MEDIATOR ШАБЛОН

- Комуникацията между обектите е отделена и централизирана чрез обекта посредник, което позволява повече гъвкавост и по-добра поддръжка.